Pre-H0 desk research on design v1.2: code audit, numerics, clip mechanism, design statistics, toy dry run (2026-10-26)

Scope and constraints

What this is. This round tests the claims and the analysis plan of `docs/op-print-design-v1.2.md` with everything that can be done **before GPU time**. Every number below is reproducible from the scripts named in each section; outputs are committed next to the scripts.

What it is not. Nothing here is a Qwen3 measurement:

- There is no GPU in this environment.
- `huggingface.co`, `arxiv.org` and `alphaxiv.org` were unreachable from both the container and the fetch tool, so:
  - no Qwen3 weights, tokenizer or dataset could be loaded;
  - OP-Print (arXiv 2610.31616) and the other cited papers could not be re-read.
- The OPSD source on GitHub *was* readable.

Statements about Qwen3 logit scales, OP-Print's own scoring settings and the Qwen3 chat template therefore remain **[unverified]**, and the logit-level experiments sweep the unknown scales instead of measuring them.

Markers follow the design: **[verified]**, **[inference]**, **[unverified]**, **[assumption]**.

Summary of findings and what they change

#	Finding	Evidence	Design change (v1.3)
F1	OPSD computes its loss at the **sampling temperature 1.1**	Code (R1)	U, P_U, m_U/r_dir and the clipped D_θ are computed at T = 1.1; Δ is scored at both T = 1 and T = 1.1
F2	The 1.7B/4B/8B scripts run a **thinking-mode teacher against a non-thinking student**. The teacher prompt also adds explicit "explore and backtrack" instructions.	Code (R1)	§1.0 adds Δ_1, the argmax-chunk index and a **no-r teacher context**. `--teacher_thinking False` becomes a secondary arm.
F3	Right-padded prompts are concatenated with the generation, so in training **t = 1 is read at a pad position** for all but the longest prompt in each micro-batch	Code (R1), [inference] about the effect	Our scorer uses no mid-sequence pads; G/G-Max are also reported without t = 1 / chunk 1
F4	The training budget is **48 optimiser steps, with 3 on-policy rollouts per member** (effective batch 32)	Code (R1)	§1.1 pins effective batch 32 and `vllm_sync_frequency`
F5	**Every clamped logit is pushed down** (g_a = q_a·P_U ≥ 0). The clipped loss has negative-loss optima that *abandon* teacher tokens with p ≳ clip, so q = p is only a local minimum.	Algebra + fp64 checks (R3)	§0.1 rewritten. H3 sign effects are expected at clip 0.05 already, not only at 0.01.
F6	v1.2's competitor **m_U = Σ_U f is negative at essentially every correction site** and does not track trainability (Spearman ≈ 0.05–0.21 with the direct first-order KL change at clip 0.05)	R3	m_U is replaced by **r_dir**, the first-order full-KL reduction under the clipped gradient. m_U is kept as secondary.
F7	The **fp32 log1p form** is 2–8× more accurate than Barron's form at V = 151,936. The fp32 direct form is unusable below KL ≈ 1e-3.	R2	log1p becomes the primary Δ implementation; Barron becomes the cross-check
F8	The bf16-logit rounding floor follows **$\Sigma_k p_k(1-p_k)\cdot s_k^2/12$**: ≈ 1e-4 / 4e-4 / 1e-3 nats/token for top logits near 8 / 16 / 24. It is **exactly 0 when the two passes give bitwise-identical logits.** The fp32 head cuts it ~10×.	R2	The reviewer's 3e-4 prior is confirmed for top logits in [8, 16). N1 is not even a reliable lower bound. N2 stays primary. A per-position analytic floor is added as a diagnostic.

#	Finding	Evidence	Design change (v1.3)
F9	The bf16 OPSD gradient has 3–8% relative error at T = 1 and 6–22% at T = 1.1 (dividing bf16 logits by T rounds them twice). At near-equal positions it is noise. At clip 1e-6, 350–700 vocabulary entries per position change clamp status.	R2	H5 motivation strengthened. Clip 1e-6 in bf16 is numerically ill-defined, so the optional 1e-6 dose is dropped unless run in fp32.
F10	The **v1.2 sign-flip placebo is anti-conservative for H1's null** when leakage is heterogeneous: 10.7–13.7% rejections at nominal 5%. HC3 OLS holds the level (4.0–7.0%).	R4 (S2)	H1's primary inference is OLS + HC3 with 16 assignment-pattern fixed effects and a spline in d. The sign-flip placebo stays as a sharp-null sanity check.
F11	Gate feasibility depends on the difficulty share f_a of score variance. A variance-based post-H0 bound on split-half r is valid but **too loose to be useful**: 0.44–0.96 in the toy where the true r ≈ 0. A **staged gate after 2 split/complement pairs** stops correctly in all 9 toy arms.	R4 (S1), R5	Staged gate: continue to 8 models only if 2r_11/(1 + r_11) ≥ 0.3
F12	Under a chunk model calibrated to OP-Print's 1.7B AUCs, H3's MDE for a gap difference is **0.007 with 8 vs 8 models and 0.016 with 2 vs 2**	R4 (S3)	The 0.2/0.01 cells are labelled "AUC-level only". The 8-model clip-0 arm stays.
F13	**Toy dry run.** Clip-off leakage drops sharply under any clip (AUC 0.576 → 0.52–0.53). gap(0) − gap(0.05) = +0.038 [0.030, 0.046]. The template spike makes G-Max a chunk-1 attack at clip ≥ 0.01. |score − median_out| is uninformative. The θ0-teacher arm is the strongest score and flips sign at clip 1e-6. Leakage is homogeneous, so r ≈ 0.	R5	H3: AUC-by-dose as a co-primary outcome, a restricted trend test, and the learned aggregator replacing |·| as the primary sign-agnostic attack. H1: one primary test per clip.

R1. OPSD source audit

Full text: `docs/verification/opsd-code-audit.md`.

Confirmed. All v1.2 code facts were confirmed against the source (clip semantics, β = 0 direction, no-grad fixed teacher, merge/unmerge without an fp32 base copy, script hyper-parameters, README statements). The "tinker-style loss is unclipped" item moves from [inference] to [verified]: it is a sampled-token policy gradient with no clamp. All of v1.2's arithmetic was recomputed and is correct.

New, and design-relevant: F1–F4 above, plus:

- **Clip semantics depend on the entry point.** `opsd_train.py` maps `jsd_token_clip ≤ 0` to `None`, but the trainer itself only checks `is not None`. Calling the trainer directly with 0.0 would therefore clamp *all* positive summands. The H3 harness must pass `None` for the clip-off arm.
- **Dataset and fields.** The data is `siyanzhao/Openthoughts_math_30k_opsd`, with fields `problem` and `solution`. `--max_length 20000` means r is not truncated.

Why F2 matters for the attack [inference]. With `enable_thinking=False`, the student's generation starts after an empty think block. The teacher's context ends at `assistant\n` and is followed directly by the student's tokens. At t = 1 the teacher therefore expects `<think>`, a token the student essentially never emits.

That gives a large Δ_1 in every sample. It is clamped at any clip ≥ 1e-6, so it is untrained. It can make G-Max select chunk 1 regardless of membership.

The instruction text ("re-derive … exploring approaches and backtracking") and the thinking mode both raise teacher probability on reflective tokens. That is a content-independent explanation for the README's "style tokens 6–15× higher KL".

Neither component is sample-specific, so both can be learned generically. They add Δ variance that carries no membership information. R5 reproduces this mechanism in a toy.

R2. Numerics at the Qwen3 vocabulary

Script and outputs. `experiments/numerics/large_vocab_numerics.py` (outputs `.json` and `.out.txt`).

Setup. Synthetic logits at V = 151,936, with a bulk of sd 2.5 and candidate tokens lifted to a top-logit level ∈ {8, 16, 24}. The level is swept because Qwen3's is unknown. P3/P4 use a synthetic V × 2048 LM head stored in bf16.

P1. OPSD's loss path, low precision vs an fp64 reference on the same rounded inputs

Median relative error of the gradient w.r.t. student logits:

regime (KL scale)	bf16, T = 1	bf16, T = 1.1	fp32
correction site (≈ 0.2)	2.8e-2	7.7e-2	≤ 6e-6
correction site, clip 0.05	7.8e-2	1.7e-1	≤ 1.1e-5
diffuse sd 0.5 (≈ 0.1)	2.5e-2	6.2e-2	≤ 1e-5
diffuse sd 0.1 (≈ 5e-3)	8.2e-2	2.2e-1	≤ 1.2e-5
near-equal (≈ 1e-5)	44× (noise)	27× (noise)	6–9e-4

Absolute errors. The bf16 gradient error is ≈ 1e-3 per position at T = 1 and ≈ 3e-3 at T = 1.1. Informative positions have gradient norms of 1–4e-2.

Why T = 1.1 is worse. `logits / T` on bf16 logits rounds them a second time.

Clip 1e-6 in bf16. 350–700 vocabulary entries per position (25–30 at near-equal positions) have a different clamp status from fp64. At clip 0.05 no flips were observed. On the teacher's top token, bf16 and fp64 give the same gradient sign in only 46–61% of near-equal positions.

P2. KL value formulas in fp32 at V = 151,936

Maximum relative error vs an fp64 log1p reference. The reference was checked against 40-digit mpmath at 2.4e-11.

logit perturbation σ (KL)	fp32 direct	fp32 Barron −Σ p s	fp32 log1p(Σ p(expm1 d − d))
1e-6 (4e-13)	2.1e6	0.31	0.066
1e-4 (3e-9)	2.3e2	4.0e-3	5.1e-4
1e-2 (3e-5)	0.18	6.3e-5	3.1e-5
0.1 (2.5e-3)	5.5e-3	5.8e-5	2.8e-5
0.5 (6.8e-2)	1.9e-4	5.6e-5	3.2e-5

The log1p form, which Appendix A did not run in fp32, is the most accurate at every scale. Both log-space forms are adequate for KL ≳ 1e-5.

P3/P4. Logit-quantisation floor

Setup. Hidden-state pairs with controlled true Δ, scored by three pipelines and compared with the fp64 head on the unrounded hidden state:

- the HF default (bf16 head, so bf16 logits);
- OPSD's bf16 `F.kl_div` on bf16 logits;
- an fp32 head on bf16-rounded hidden states.

Results.

- **bf16 logits.** The mean error at true $\Delta > 0$ is of the order of the analytic prior $\Sigma_k p_k(1-p_k)s_k^2/12$, where s_k is the bf16 spacing at z_k. That is ≈ 1.3e-4 (top logit ≈ 8), ≈ 4e-4 (≈ 16) and ≈ 9e-4 (≈ 24).
  - The reviewer's ≈ 3e-4 is right for top logits in [8, 16) and about 3× too low if Qwen3's top logits sit in [16, 32).
  - Positive bias is clearest when true Δ is small. At larger Δ the per-position error is dominated by a zero-mean cross term.
- **OPSD's bf16 `F.kl_div`** errs by ±1e-3 with no consistent sign.
- **fp32 head** errs by ≈ 1–4e-5 at small Δ and ≈ 1e-4 at Δ ≈ 0.02, about 10× lower. This includes only the *final* hidden-state rounding, not accumulated trunk error, so N2 is still required.
- **Bitwise-identical logits.** At true Δ = 0 with identical hidden states, every pipeline gives **exactly zero**: identical inputs round identically.
  - Any N1-style test whose two passes are bitwise identical therefore measures nothing.
  - The floor only appears once the two contexts' logits differ by a fraction of the spacing, which is always the case in OPSD.
  - This supports v1.2's choice of N2 (an fp32 reference forward pass) as primary.
- **Aggregated to 64-token chunks (P4).** The relative bias of G's chunk mean is:
  - +44–55% (bf16 head) vs +5–10% (fp32 head) at Δ ≈ 3e-4;
  - +5–36% vs 0.4–2% at Δ ≈ 2e-3;
  - ≤ 6% vs ≤ 0.4% at Δ ≈ 2e-2. G-Max's bias is similar.

Implication. With the fp32 head, the floor matters only if member Δ falls below ≈ 1e-3 nats/token. A cheap per-position diagnostic, $\hat{\varepsilon}_t = \Sigma_k p_k(1-p_k)s_k^2/12$, can be logged by the scorer.

R3. What the element-wise clip does (beyond §0.1)

Script and outputs. `experiments/clip_mechanism/clip_mechanism.py` (fp64; outputs `.json` and `.out.txt`).

M1: clamped logits are pushed down. For any clamped entry a ($f_a >$ clip), $g_a = q_a \cdot P_U \geq 0$, so gradient descent never raises a clamped logit. Confirmed on 300 random positions at each of 4 clips: the minimum of g over clamped entries is 0.

M2: the clipped objective has negative-loss optima away from the teacher. With free logits:

- $\inf L = \min_C [\ |C| \cdot \text{clip} + h(P_C)\ ]$, with $h(P) = (1-P)\cdot\log(1-P)$, which has its minimum $-1/e$ at $P = 1 - 1/e$;
- C is the "abandoned" set: $q \to 0$ on C and $q \propto p$ elsewhere;
- one token with $p_k - p_k^2/2 >$ clip already gives $L < 0 = L(q = p)$.

So $q = p$ is only a local minimum.

- **Worked example (v1.2 §0.1).** Gradient descent from the uniform student reaches $L = -0.1802$ (the bound is -0.1803), with **q_a = 9e-4: the teacher's 0.9-probability token is abandoned.** The full KL rises from 2.38 to 5.95. Started near $q = p$, gradient descent stays at $q = p$.
- **300 random positions (V = 50).** Long gradient descent never beats the convex lower bound. It ends with $L < 0$ in 18 / 72 / 96 / 100% of runs and **with the full KL above its starting value in 34 / 67 / 87 / 98%**, at clip 0.2 / 0.05 / 0.01 / 1e-6.

M3: one clipped step can raise the full KL. The first-order change of the full KL under one clipped step is $-r$, with $r = \langle q - p, g \rangle$. $r < 0$ means that step *raises* the full KL. In the worked example $r = -0.0045$ at every clip ≤ 0.2.

M5: the closed form for the clamped token's direction. For clamped a:

$d \log q_a \propto P_U \cdot (\Sigma_i q_i^2 - q_a) - \Sigma_{i \in U} p_i q_i$

It matches direct computation in 288/288 cases. q_a **rises when the student is sharp** (entropy ≤ 2 nats over V = 1000) and **falls when it is diffuse** (≥ 4 nats). That is v1.2's "can move the wrong way at high student entropy", made precise. Its magnitude is small either way: the logit itself is always pushed down (M1).

M6: synthetic ensemble at V = 151,936.

T	clip	regime	share of positions with r < 0	share with m_U < 0
1.0	0.05	correction site	80%	100%
1.1	0.05	correction site	82%	100%
1.0	0.05	diffuse	38%	52%
1.1	0.05	diffuse	20%	28%
1.1	0.01	diffuse	68%	62%
1.1	1e-6	any	30–100%	100%

- **Temperature changes the clamped set substantially.** For diffuse positions at clip 0.05, the median clamped share of KL is 105% at T = 1 and **0% at T = 1.1**.
- **m_U tracks r poorly.** Spearman rank correlations with r at clip 0.05 are m_U +0.05 / +0.21 (T = 1 / 1.1), m_U^+ (positive unclamped mass) +0.22 / +0.36, and the clipped loss +0.40 / +0.45. At smaller clips they turn negative.

Implications.

1. **H1's mechanistic competitor.** v1.2's m_U is not a mass: it is negative wherever the teacher's preferred token is clamped. The proposed replacement is $r_{dir}(x) = \text{mean}_t \langle q_t - p_t, g_t \rangle$ at T = 1.1. It is the first-order reduction of the *full* θ0-teacher KL that one clipped step would make directly, under an isotropic-NTK **[assumption]**. m_U and m_U^+ are kept as secondary.
2. **H3 sign effects.** At clip 0.05, the direct effect at the high-Δ sites that G-Max reads points *away* from the teacher. Members can therefore end up with *higher* Δ at those sites. Sign-agnostic attacks are needed at clip 0.05, not only at 0.01.
3. **OPSD's own clip of 1e-6.** The README's 1e-6/1e-7 setting optimises an objective whose minimisers concentrate the student on a set of teacher mass 1/e. A sharpening, entropy-minimisation-like pressure **[inference]**. This bears on interpreting the "non-thinking" results. It is not needed for the minimal paper.

R4. Design statistics

Script and outputs. `experiments/design_sim/design_sim.py` (outputs `.json` and `.out.txt`).

Model. The per-model log score is $l_m(x) = a(x) + g_m - s_m \cdot b(x) \cdot l_m(x) + e_m(x)$, on the 4 splits + complements design ($N_+ = 1000$).

S1. Reliability gate

Closed forms: $AUC = \Phi(\mu_b / \sqrt{2\sigma_a^2 + \tau_b^2 + 2\sigma_e^2})$ and $r_{raw} = \tau_b^2 / (\tau_b^2 + \sigma_e^2)$. Monte Carlo agrees: r = 0.723 vs 0.723 and 0.203 vs 0.204, with AUC 0.594 reproduced.

At the paper's G-Max AUC of 0.594, the raw split-half r (8-model Spearman–Brown in brackets) is:

difficulty share f_a	CV(leakage) 0.5	1	2
0.5	0.06 [0.10]	0.20 [0.34]	0.62 [0.77]
0.8	0.13 [0.23]	0.44 [0.61]	infeasible
0.9	0.25 [0.40]	0.72 [0.84]	infeasible

difficulty share f_a	CV(leakage) 0.5	1	2
0.95	0.44 [0.61]	infeasible	infeasible

Reading. The gate (raw r ≥ 0.3) passes only if sample difficulty dominates the in-vs-out score variance, or leakage is strongly heterogeneous.

Post-H0 bound (new). Four inputs, all available before the 7 further models are trained:

- $\sigma_tot^2 = Var(\ell \mid in) + Var(\ell \mid out)$, on H0's model (under normality this equals $(\hat{\mu}_b/\Phi^{-1}(AUC))^2$, with $\hat{\mu}_b$ the mean out − in difference of log Δ);
- σ_a^2 from θ0's between-sample variance of log Δ, net of rollout noise. This assumes training leaves the difficulty variance unchanged **[assumption]**;
- σ_roll^2, the rollout part of σ_e, from two independent K = 8 rollout sets per sample;
- $R = \sigma_tot^2 - 2\sigma_a^2$.

Then $r_raw \le (R - 2\sigma_roll^2)/(R - \sigma_roll^2)$, because $\sigma_e \ge \sigma_roll$.

This bound is valid but loose, so it is not used as the gate. When rollout noise is a small part of σ_e it barely binds: 0.88 against a true 0.20 in the table above. In the toy (R5) it gave 0.44–0.96 where the true r ≈ 0.

v1.3 therefore uses a staged gate. After 2 split/complement pairs (4 models), compute the split-half r_11 across the pairs (1 in / 1 out per half) and continue only if the Spearman–Brown projection $2r_11/(1 + r_11) \ge 0.3$. That is the same quantity the 8-model gate tests, at half the cost.

S2. H1 inference

Rejection rates at nominal 5%, one-sided, 300 replications each (MC SE ≈ 0.013):

scenario (H1 null unless stated)	v1.2 sign-flip placebo	v1.1 any-4-of-8	OLS HC3	HC3 + pattern FE + spline(d)	Freedman–Lane
sharp null (no leakage)	0.047	0.050	0.043	0.047	0.050
homogeneous leakage	0.057	0.060	0.067	0.063	0.063
heterogeneous leakage (CV 1.5)	**0.107**	**0.127**	0.040	0.040	0.043
+ model offsets/scales	**0.127**	**0.140**	0.070	0.070	0.077
+ heteroscedastic noise	**0.083**	**0.090**	0.047	0.043	0.063
+ leakage nonlinear in d	**0.137**	**0.147**	0.047	0.047	0.070
alternative β_c = 0.15	0.883	0.893	0.770	0.753	0.797

Why the placebo fails. The within-pair sign flip is exact for the *sharp* null "membership has no effect". H1's null is different: "c adds nothing to d", with leakage present. Under sign flips, a sample's leakage b(x) enters only as ±b(x), with variance contribution $b^2/4$. The real v carries the leakage heterogeneity that is unrelated to c. When CV(b) ≳ 0.6, the placebo distribution is too narrow.

Recommendation. H1's primary test is OLS with HC3 standard errors, 16 assignment-pattern fixed effects (which absorb model-level offsets exactly) and a pre-registered cubic spline in d. Freedman–Lane is a robustness check. The sign-flip placebo is kept only as the "c does not predict v_placebo" sanity check.

S3. G vs G-Max and the H3 gap

Setup. A chunk-level model: 16 chunks, K = 8, leakage on a sample's top-2 chunks ("sparse") or on all chunks ("diffuse"). It is calibrated to OP-Print 1.7B: the sparse arm gives G 0.583 and G-Max 0.603, against the paper's 0.579 and 0.594.

Gap and its noise.

- Sparse imprint: gap +0.0195. Diffuse imprint, matched on G: gap −0.0073.
- SD of the gap is 0.0055 for one model and 0.0017 for the mean of 8 models.
- Within one model, OP-Print's 0.015 gap would then be ≈ 2.7 SD. That holds only if G and G-Max are as correlated as in this model. Paired DeLong on the real scores decides.

H3 MDE (80% power, two-sided 5%). 0.0068 with 8 vs 8 models and 0.0156 with 2 vs 2. The 0.2/0.01 cells can detect only gap changes of ≳ 0.016.

Chunk-1 spike. A membership-independent spike in chunk 1 lowers G-Max's AUC (0.602 → 0.594) only once it is ≈ 10× the median chunk Δ. Dropping chunk 1 restores it (0.601). Whether Qwen3's Δ_1 is that large is an empirical question for §1.0.

R5. Toy OPSD dry run

Scripts. `experiments/toy_opsd/`:

- `toy_opsd.py` (simulator), `run_all.sh` (all arms), `analyze.py` → `results.md`/`.json`;
- `imprint_by_class.py`, `gate_bound.py`, `staged_gate.py`;
- raw per-model scores in `runs/` and `runs_lowcap/`.

What it is. A small, fully specified system with OPSD's *mechanics*. It is **not** a model of Qwen3. Its purpose is (a) to execute the pre-registered analysis end to end before GPU time, and (b) to check that the mechanisms in R1/R3 survive in a shared-parameter network trained on-policy.

Setup

Base model and world.

- θ0 is an MLP language model (V = 64, 128-token sequences) pre-trained on a synthetic process with three components.
- **Correction sites.** After a trigger token comes a sample-specific answer. Answers are partly predictable from the question (64-dim features, κ = 1). With r the teacher gives the correct answer probability 0.88. The student gets 0.27 (chance 0.125).
- **Instruction asymmetry.** A "reflective" flag raises style-token mass from 0.03 to 0.12.
- **Template asymmetry.** A "thinking" flag makes the first token THINK, with probability 0.98.
- Per-token θ0 KL between the contexts: 6.0 nats at t = 0, 1.18 at answer positions, 0.09 elsewhere.

OPSD training.

- LoRA rank 16 on every linear layer (α = 2r).
- On-policy rollouts at T = 1.1, top-k 20, top-p 0.95.
- OPSD loss at T = 1.1 with an element-wise clamp.
- Adam, max grad norm 0.1, 32 prompts × 1 rollout per step, 3 epochs = 48 steps.

Design and scoring.

- 4 splits + complements (8 models per arm), with the same splits in every arm.
- K = 8 scoring rollouts, 16-token chunks (8 per rollout).

Calibration and arms.

- **lr = 3e-3** was chosen so that the clip-off arm matches OP-Print's AUC level. This uses the attack AUC, which the real study forbids; it is a toy-only calibration.

- Arms:
  - clip ∈ {off, 0.2, 0.05, 0.01, 1e-6};
  - no-template ("nothink") at {off, 0.05};
  - lr 3e-4 at {off, 0.05}. This is the smallest lr meeting the design's f_rm ∈ [0.2, 0.5] rule at clip 0.05;
  - a low-capacity configuration (rank 4, 16-dim questions) at clip off.

Results

AUC by arm (mean of 8 models; bootstrap 95% CI):

arm	f_rm (train)	G	G-Max	G-Max w/o chunk 1	learned aggregator	H1′ S_θ − S_θ0	H1′ D_θ (θ0 teacher)	|G − med_out|
clip off	+0.68	0.576 [0.568, 0.583]	0.608 [0.599, 0.617]	0.606	0.630	0.620	**0.640**	0.489
clip 0.2	+0.42	0.533	0.526	0.542	0.558	0.551	0.554	0.497
clip 0.05	−0.41	0.523 [0.515, 0.530]	0.517 [0.508, 0.528]	0.532	0.536	0.536	0.536	0.494
clip 0.01	−6.3	0.525	0.511	0.527	0.548	0.544	0.517	0.495
clip 1e-6	−83	0.502	0.501	0.503	0.500	0.504	**0.480 [0.469, 0.489]**	0.507
nothink, off	+0.59	0.579	0.613	0.611	0.640	0.627	0.643	0.491
nothink, 0.05	−0.55	0.521	0.532	0.530	0.541	0.534	0.530	0.491
lr 3e-4, off	+0.64	0.517	0.528	0.529	0.548	0.530	0.539	0.497
lr 3e-4, 0.05	+0.45	0.506 [0.501, 0.511]	0.508	0.508	0.520	0.512	0.510	0.504
low capacity, off (lr 3e-4)	+0.56	0.505	0.506	0.507	0.501	0.508	0.510	0.504

H3 gap (G-Max − G):

arm	gap	without chunk 1
clip off	+0.032 [0.028, 0.037]	+0.033
clip 0.2	−0.007 [−0.015, 0.001]	+0.012 [0.007, 0.016]
clip 0.05	−0.006 [−0.014, 0.003]	+0.011 [0.006, 0.015]
clip 0.01	−0.015 [−0.023, −0.006]	−0.001
clip 1e-6	0.000	0.000

Paired comparisons (same samples and splits):

- gap(off) − gap(0.05) = **+0.038 [0.030, 0.046]**;
- without the template: +0.023 [0.018, 0.029];
- at lr 3e-4: +0.008 [0.001, 0.015].

The Spearman trend over the 5 doses is −0.3.

Where the imprint lands (`imprint_by_class.py`; model 0 of each dose; log(mean Δ_out / mean Δ_in) by position class):

clip	answer positions (4.6%)	t = 0	style / other	log q(correct) θ − θ0 at answer positions, members / non-members
off	**+0.174**	+0.105 ($\Delta_1 \approx 0.04$)	−0.007 / −0.006	**+0.16** / −0.11

clip	answer positions (4.6%)	t = 0	style / other	log q(correct) θ − θ0 at answer positions, members / non-members
0.2	+0.049	−0.005 ($\Delta_1 \approx 5.4$)	−0.001 / −0.001	−0.06 / −0.12
0.05	+0.036	−0.009 ($\Delta_1 \approx 5.2$)	+0.002 / 0.000	−0.04 / −0.10
0.01	+0.017	−0.013	style share collapses 0.115 → 0.001	−0.05 / −0.09
1e-6	(no answers emitted)	—	99% "other" tokens	—

§1.0 diagnostic.

arm	argmax chunk is chunk 1	argmax token is an answer
clip off	8%	99%
clip ≥ 0.01 (template on)	97–98%	2–4%
nothink	9%	99%

The student emits THINK first in 98.5% of rollouts at clip off and in ≤ 0.2% at any clip.

H1 and gates.

- **Reliability.** Split-half reliability of v is −0.05 to +0.05 in every arm. The ≈ 30 competitor tests produce a few |t| ≈ 2–2.9 by chance.
- **Variance-based post-H0 bound.** Valid but loose: 0.44–0.96, and undefined in two arms where θ0's difficulty variance exceeds the trained model's total variance.
- **Staged gate.** Correct "stop" in all 9 arms after 4 models: projected $r_{22} \leq 0.09$.

Lessons for the design

1. **The clip strongly shapes leakage itself, not only the G-Max − G gap.**
   - Any clip ≤ 0.2 cuts the clip-off AUC from 0.58–0.61 to 0.51–0.53. At 1e-6 it is 0.50.
   - H3 gains a co-primary outcome: the AUC level of the strongest pre-registered attack as a function of dose, reported with utility. This is the "clip as an empirical defense" question.
   - In OP-Print the clip-0.05 AUC is 0.59, much higher than the toy's 0.52. Real correction sites probably include many entries whose summands sit below 0.05, or indirect routes are stronger at LLM scale **[inference]**.
2. **The sparse-imprint prediction of H3 holds in the toy.** The membership imprint sits at the correction sites (answer positions +0.174), and the gap widens as clip → 0 (+0.038). The paired-bootstrap primary contrast has ample power at this effect size.
3. **The template spike can hide the G-Max advantage.**
   - At clip ≥ 0.01 the THINK mismatch is not learned. $\Delta_1 \approx 5$ nats, and chunk 1 is the argmax in 97–98% of rollouts.
   - G-Max then reads only t = 1 and is no better than G: the gap is −0.006 to −0.015, significant only at 0.01. Dropping chunk 1 restores a significant positive gap (+0.011 at clip 0.05).
   - Unclipped, the student *learns to emit the teacher's template token*, in 98.5% of rollouts. This supports the §1.0 additions and the template arm. It also suggests one concrete way the clip "stabilises" OPSD with the default thinking-teacher setup **[inference]**.
4. **Trend tests across doses mislead when leakage vanishes.** Spearman = −0.3 here, because clip 1e-6 has no leakage at all. The primary stays the paired gap(0) − gap(0.05). The trend is secondary, restricted to doses whose G or G-Max AUC CI excludes 0.5, and also reported for gap/(AUC_G − 0.5).
5. **R3's anti-teacher dynamics survive in a network.**

    - The θ0-teacher residual on training samples *grows* under clipping: f_rm +0.68 / +0.42 / −0.41 / −6.3 / −83 from clip off down to 1e-6.
    - The correct-answer probability falls for all samples under any clip.
    - At clip 0.01 the student stops emitting style tokens, whose summands of ≈ 0.02 lie between 0.01 and 0.05. At 1e-6 it collapses.
    - Membership then shows with the **opposite sign** in the training-objective arm: D_θ AUC 0.480 [0.469, 0.489]. AUC < 0.5 is to be reported, not folded.
6. **|score − median_out| is not a usable sign-agnostic attack.** It is uninformative in every arm, including where G gives 0.58, because it detects dispersion, not a location shift. The learned aggregator on the sorted CRN chunk vector is the sign-agnostic attack that works (≥ G-Max in all clipped arms). Two-sided AUC CIs cover single scores.
7. **H1′ arm 3 (θ0 teacher, D_θ) is the strongest single score** at clip off (0.640, vs OP-Print's G-Max 0.608). Δ_θ − D_θ has AUC < 0.5 (0.451). The adapter's effect on the teacher side works against OP-Print's Δ_θ, which therefore under-uses the signal.
8. **f_rm is not a proxy for leakage.**
    - The f_rm-rule lr (3e-4) gives f_rm +0.45 at clip 0.05 with barely detectable leakage (G 0.506 [0.501, 0.511]).
    - The low-capacity configuration has f_rm +0.56 with no leakage (0.505).
    - The template and style components are learned generically.
    - Supports the f_rm^r (no-r) caveat in v1.3 §1.1.
9. **AUC ≈ 0.6 does not imply a testable H1.** Leakage in the toy is homogeneous across samples, so v has no reliable per-sample variance. The gate correctly stops H1.
    - The variance-based post-H0 bound is too loose to act as a gate; the staged gate after 2 pairs works.
    - With ≈ 30 secondary tests, chance |t| ≈ 2–3 appear. H1 keeps **one primary test per clip** (c at clip 0 and at clip 0.05, with Holm correction). Competitors are secondary.

What the toy cannot show.

- Qwen3's real clamp structure.
- LLM-scale spillover.
- Whether style tokens dominate the argmax: the toy's style Δ is diffuse and never the maximum.
- The real step size: the toy lr is calibrated on AUC.

The toy validates the analysis code and the mechanisms. It does not predict Qwen3's numbers.